

WPE High-Level Practice Challenges

5 Elite Problems with Progressive Hints & Niche Shortcuts

Designed for IOQM/INPhO Prep — September 2026

This practice sheet has been engineered to stretch your physical and mathematical intuition using the exact concepts from your first four coaching sessions. In accordance with your request, **no final numerical answers or worked-out algebraic steps are provided**. Instead, each challenge is accompanied by:

1. **The Setup:** A rigorous Olympiad-level mechanical description.
2. **The Olympiad Niche Shortcut:** A theoretical framework that collapses complex integrals or multi-body systems into simple, elegant relations.
3. **Progressive Hints:** Step-by-step conceptual nudges designed to help you construct the final equations on your own without spoiling the mathematical breakthrough.

Challenge 1: Wedge Frame & Galilean Invariance

The Setup:

A smooth wedge of mass M and inclination angle θ sits on a completely frictionless horizontal floor. A block of mass m is placed on the wedge. An external horizontal force F is applied to the wedge, accelerating it horizontally to the right with a constant acceleration $a = g \tan \theta$, which keeps the block completely stationary relative to the wedge.

1. Find the work done by the normal contact force $\vec{N}$ on the block alone in the ground frame (S) during a horizontal displacement s of the wedge.
2. Now, look at the system from a frame (S') moving horizontally to the left with a constant velocity u relative to the ground. Determine the work done by the normal contact force $\vec{N}$ on the block alone in this moving frame during the time interval t (where t is the time taken to travel distance s in the ground frame).
3. Prove that the net internal work done by the action-reaction normal forces between the wedge and the block is zero in both frames, demonstrating the Galilean Invariance of Net Internal Work.

Challenge 2: The Sinusoidal Track and Friction Projection

The Setup:

A small block of mass m is slowly hauled by a tangential pulling force $\vec{F}$ along a rough curved track. The vertical profile of the track is given by the sinusoidal function:

$$y(x) = h \sin^2 \left(\frac{\pi x}{2L} \right) \quad \text{for } x \in [0, L]$$

The coefficient of kinetic friction between the block and the track is μ . Find the total work done by the pulling force $\vec{F}$ in moving the block from the bottom of the track ($x = 0$) to the top end of the track ($x = L$) with a negligible, constant speed ($v \approx 0$).

Challenge 3: Conservative Fields & Topological Singularities

The Setup:

A force field in the XY -plane is defined at all points except the origin $(0,0)$ by:

$$\vec{F}(x,y) = \frac{-y\hat{i} + x\hat{j}}{x^2 + y^2}$$

This vector field describes a singular vortex.

1. Evaluate the 2D curl of the field: $\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}$ at any point $(x,y) \neq (0,0)$. Does it pass the mathematical test for a conservative force field?
2. Calculate the work done by this force along a closed circular path of radius R centered at the origin, starting and ending at $(R,0)$, moving once counter-clockwise.
3. If the curl is zero, why is the closed-loop work non-zero? Explain the topological restriction of conservative fields.

Challenge 4: Sliding Chain and Continuous Mass-Weighted Friction

The Setup:

A heavy, uniform, highly flexible chain of total mass M and length L is draped over a double-incline with meeting angles α and β . Initially, a length x_0 of the chain rests on the rough incline of angle α (coefficient of kinetic friction μ) and the remaining length $L - x_0$ hangs down the smooth incline of angle β . The chain is released from rest. As it slides, the hanging portion on the smooth incline pulls the remaining chain over the peak. Find the total work done by the friction force on the chain by the time the entire chain has slid completely onto the smooth incline.

Challenge 5: Wrapping String and Spiral Kinematics

The Setup:

A rigid cylinder of radius R is fixed vertically on a frictionless horizontal table. A light, inextensible string of length L has one end attached to the cylinder and the other end tied to a particle of mass m . Initially, the string is fully unwound, straight, and taut. The particle is projected with an initial velocity v_0 in a direction perpendicular to the string, causing it to begin wrapping around the cylinder.

1. Show that the work done by the tension force in the string on the particle is strictly zero at every instant during the motion.
2. Using the work-energy relation, find the speed of the particle as a function of the remaining unwound string length l .
3. Calculate the tension in the string as a function of the remaining unwound string length l .